Design Your Own **Portable Spectrum Analyser**

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This low-cost portable spectrum analyser can be used to analyse the frequency content of input signals up to 5kHz. Frequency spectrum of an input signal possesses vital information, and the same has been employed for various applications ranging from engineering to biomedical, such as speech recognition, electrocardiogram (ECG) signal analysis, fault diagnosis of induction motors, music analysis and so on.

Analysing the spectra of electrical signals provides information about dominant frequency, power, distortion, harmonics, bandwidth and other spectral components, which are not easily detectable in time-domain waveforms. Spectrum analysers are quite expensive and might not be affordable for individuals such as students and electronics hobbyists.

The block diagram of the proposed low-cost portable spectrum analyser is shown in Fig. 1. Snapshot of the authors' prototype using Arduino DUE board and test setup used for evaluation is shown in Fig. 2.

The system employs Fast Fourier Transform (FFT)-based frequency analysis. FFT is a computationally efficient algorithm for computing Discrete Fourier Transform (DFT) of sample sizes that are positive integer powers of two.

ADC Sampling Frequencies on Arduino Due — Table 1

Pre-Scaler Value	Samples/ Second	ADCSRA Register Value
2	6,15,385	001
4	3,07,692	010
8	1,53,846	011
16	76,923	100
32	38,462	101
64	19,231	110
128	9,615	111

Frequency Resolution for Different Parameters — Table 2

N → / f_s ↓	16	32	64	128	256	512	1024	2048
6,15,385	38400	19200	9600	4800	2400	1200	600	300
3,07,692	19200	9600	4800	2400	1200	600	300	150
1,53,846	9600	4800	2400	1200	600	300	150	75
76,923	4800	2400	1200	600	300	150	75	37.5
38,462	2400	1200	600	300	150	75	37.5	18.75
19,231	1200	600	300	150	75	37.5	18.75	9.375
9,615	600	300	150	75	37.5	18.75	9.375	4.6875

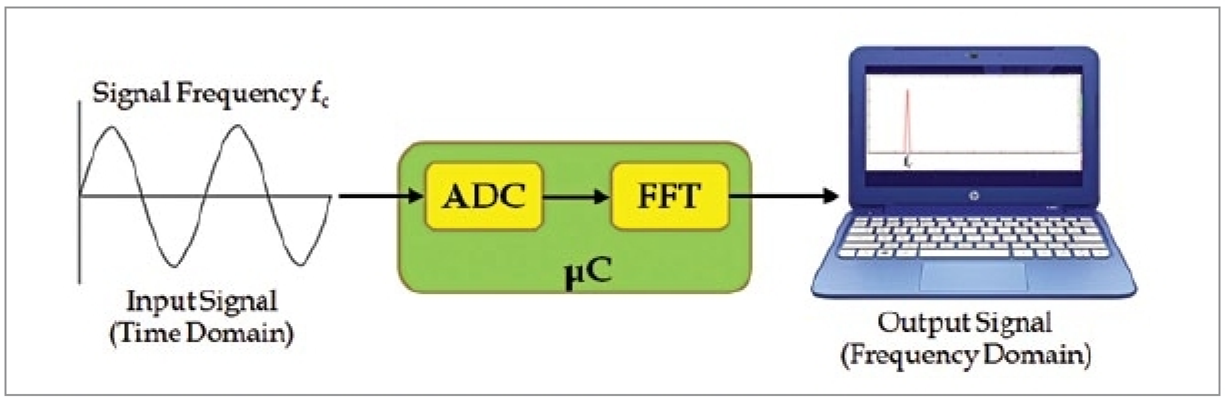

Fig. 1: Block diagram of the low-cost portable spectrum analyser

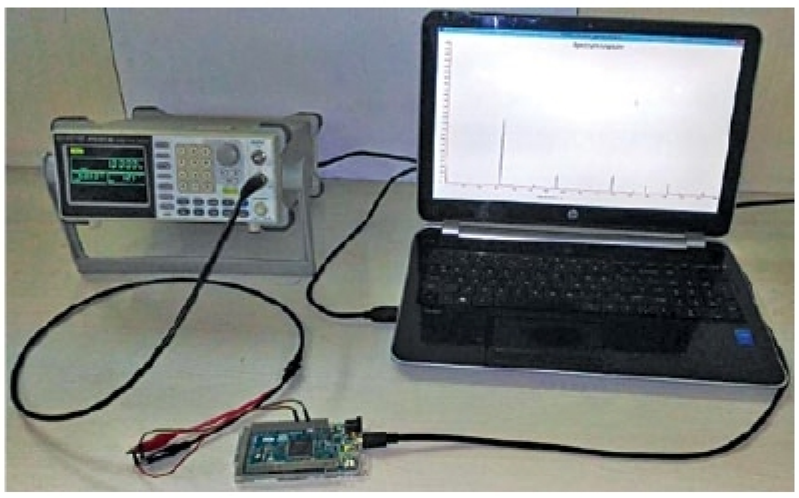

Fig. 2: Authors' prototype

Table 1 lists the possible sampling frequencies available with Arduino Due board. These frequencies are selected using a pre-scaler value in ADCSRA register of an on-board analogue-to-digital converter (ADC). Resolution of the frequencies can be increased by either reducing the sampling frequency (f_s) or by increasing the number of FFT points (N). Based on frequency resolution requirements of an application, suitable values can be selected by the designers. Table II lists the frequency resolutions for different combinations of N and f_s.

A function generator is used to evaluate a system by providing input signals at different frequencies and monitoring the variation in the frequency spectrum on the laptop. An input signal is given to one of the ADC channels available onboard. Arduino

EFY Note
The source code of this project is available for free download at *source.efymag.com*